

Skills

Test & Validation:

HIL rigs (Vector VT System, RT Rack), Fault Insertion Unit (FIU), CANoe/CANalyzer, LabVIEW TestStand

Embedded Programming:

C/C++, Python, Shell

Automotive Protocols:

UART, CAN, RS-232, RS-485, UDS

AUTOSAR & Config Tools:

DaVinci, EB Tresos, ARXML files

CI/CD Pipeline Engineering:

Jenkins (Groovy, declarative pipelines), GitHub Actions, Git/Gerrit/GitLab, Jira, Continental ATP/TEMPPO, pipeline-as-code triggers for HIL/SIL regression runs

Test Reporting & Dashboards:

JUnit/Robot Framework/TestStand result parsing, Allure Report, Grafana/Kibana pass-fail and flaky-test trend dashboards, Docker-based SIL runners

Debuggers:

Lauterbach/Trace32, JTAG, PE micro MULTILINK, ICAS1 Debugger

Configuration Management:

PTC Integrity, IBM DOORS, MKS

Standards & Process:

ASPICE (HWE.2, HWE.3, SWE.5), ISTQB Foundation, Agile/Scrum

Cross Compilers/IDE:

ARM GCC, GHS, S32 Design Studio, Eclipse, Visual Studio

Vamsi Krishna Sai Kavety

HIL/SIL TEST ENGINEER | EMBEDDED SOFTWARE INTEGRATION

Contact

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Personal Information

- **Nationality:** German
- **Education:** Master of Science - Automotive Software Engineering — TU Chemnitz, Germany, 2018; PG Program - Data Science & AI (pursuing) — ExcelR, India, 2024-Present
- **Languages:** English - C1, German - B1
- **Driving License:** Class B/BE

About Me

- 11+ years across embedded hardware/software testing, HIL verification, and automotive module integration for VW, Daimler, and Fiat/Chrysler OEM programs.
- HIL test engineer: maintained and upgraded Vector VT System/RT Rack rigs, executed Fault Insertion Unit (FIU) testing, and automated regression suites with LabVIEW TestStand and Python.
- Automotive software integrator: Gateway Control Unit (ICAS1/ID.3) and Telematics/Body Control module integration using AUTOSAR (DaVinci, EB Tresos) with Jira/Jenkins CI/CD.
- ASPICE-aligned: implemented HWE.2, HWE.3, SWE.5 process phases; hands-on with Continental ATP/TEMPPO test management framework.
- Test automation & reporting: builds Jenkins/GitHub Actions pipelines that trigger HIL/SIL regression on every commit, parses JUnit/Robot Framework/TestStand result files into structured reports, and publishes pass/fail and flaky-test trend dashboards (Grafana/Kibana); pursuing a PG program in Data Science & AI to extend failure-pattern analysis.

Core Competencies

- HIL/SIL Test Engineering: Vector VT System, RT Rack, CANoe/CANalyzer, LabVIEW TestStand, Fault Insertion Unit (FIU), root cause analysis.
- Automotive Software Integration: Gateway/Telematics/Body/Powertrain ECUs, AUTOSAR (DaVinci, EB Tresos, ARXML), software flashing & diagnostics (ODIS).
- Test Automation & CI/CD: Jenkins/GitHub Actions pipeline authoring (Groovy, pipeline-as-code) triggering HIL/SIL regression, Python-based regression and fault-injection scripting, Git/Gerrit/GitLab, Jira, Continental ATP/TEMPPO.
- Debugging & Diagnostics: Lauterbach/Trace32, JTAG, PE Micro Multilink debuggers, CAN/UART/RS-232/RS-485 protocol analysis.
- Test Reporting & Dashboards: JUnit/Robot Framework/TestStand result parsing, Allure Report, Grafana/Kibana pass-fail and flaky-test trend dashboards, Docker-based SIL runner environments.
- Standards & Process: ASPICE (HWE.2, HWE.3, SWE.5), ISTQB Foundation, Agile/Scrum, IBM DOORS, PTC Integrity/MKS configuration management.

Certifications

- ISTQB Certified Tester - Foundation Level (CTFL)

Current Role Focus

- HIL/SIL Test Engineer - Automotive ECU & Gateway Platforms
- Software Integration Engineer - AUTOSAR/Gateway/Telematics Systems
- Test Automation Engineer - CI/CD Pipelines, Test Reporting & Dashboards

Work Experience

Software Engineer/Test Engineer - HIL (VW)

[Manu Dextra / Vitesco Technologies - Regensburg](#)

Jul 2021 - Mar 2024

- Maintained and upgraded HIL test rigs (Vector VT System, RT Rack) across multiple ECU programs, minimizing test bench downtime through advanced hardware/software troubleshooting.
- Designed and executed fault injection tests using the Fault Insertion Unit (FIU) to simulate electrical failures and verify ECU diagnostic and fault-handling behavior.
- Automated regression and validation suites using LabVIEW TestStand and Python scripts for signal simulation and fault injection.
- Built Jenkins pipelines that triggered HIL regression runs on every ECU build, parsed TestStand/JUnit-style result files, and published pass-fail and flaky-test trend dashboards (Grafana) for the wider test team.
- Executed and validated software downloads (SWDL) on EMS ECUs, coordinating with development and test teams to ensure test environment readiness.
- Managed hardware inventory, calibration schedules, and system documentation; contributed to continuous improvement of HIL methodologies and tooling.

Software Engineer - Vehicle Testing

[Manu Dextra / Continental Automotive - Regensburg](#)

Aug 2019 - Jun 2021

- Conducted manual and automated in-vehicle/lab testing on the VW ID.3 platform to validate Gateway Control Unit (ICAS1) requirements for every sprint release.
- Owned software module integration within an Agile team (Jira) for the VW Gateway Control Unit, coordinating with cross-site development teams.
- Established and managed Jenkins-based CI/CD integration testing setups, improving release traceability and regression coverage.
- Parsed automated test-run outputs into structured pass-fail reports and tracked sprint-level regression trends in Jira, giving the team early visibility into recurring failures.
- Performed software flashing and diagnostics on hardware using ICAS1 debuggers, VW ODIS tools, and custom Python scripts.

Software Engineer/Test Engineer - ECU Verification

[IPN Brainpower / Continental Engineering Services - Nurnberg](#)

Feb 2018 - Apr 2019

- Executed automated and manual HIL test procedures covering nominal, boundary, and complex powertrain control scenarios.
- Designed fault injection tests (FIU) to simulate electrical failures and verify diagnostic/fault-handling behavior of ECUs.
- Performed data analysis and root cause analysis (RCA), reporting defects via Jira for software/hardware anomalies.

Master Thesis Candidate

[Conti Temic Microelectronics GmbH \(Body & Security\) - Markdorf](#)

Jul 2017 - Dec 2017

- Researched Over-the-Air (OTA) software/firmware update delivery mechanisms for automotive service center systems.

Student Intern - Testing

[ADC Automotive Distance Control Systems GmbH - Lindau/Ulm](#)

Jan 2017 - Jun 2017

- Developed Python scripts to automate repetitive vehicle component testing and validation tasks in line with automotive standards.

Senior Hardware Engineer / Senior HIL Test Engineer - Power Train

[Robert Bosch Engineering - Coimbatore, India](#)

May 2011 - Mar 2014

- Led integration and validation of ECUs for diesel/gasoline powertrains, coordinating with Bosch GmbH and OEMs (Fiat/Chrysler) on hardware specifications.
- Conducted functional, EMC, and environmental testing per automotive quality standards; performed root cause analysis for system integration issues.

Engineer R&D

[McML Protection Technologies - Hyderabad, India](#)

Jul 2009 - Apr 2011

- Designed proof-of-concept hardware prototypes and developed/validated embedded C/C++ firmware and drivers.
- Set up test benches and performed unit/integration/system-level testing and debugging of embedded hardware/software.